

BAYESIAN ANALYSIS OF SURVIVAL DATA USING THE RAYLEIGH MODEL AND LINEX LOSS

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Summary: Based on right censored data this paper presents the Bayes estimators for some lifetime distribution parameters such as the mean survival time, the hazard function and the survival distribution function, using the non-informative and informative prior distributions for θ . These estimators are derived using both the squared error loss function and Varian's asymmetric linear-exponential (linex) loss function. A hierarchical approach is followed when the hyperparameters of the conjugate gamma prior are unknown, which results in the use of the F-distribution as a prior for θ . A practical example illustrates the use of the non-informative and F priors under the two loss functions, as well as the prediction of the lifetime of a future observation/patient, for the case of unknown hyperparameters.

1. Introduction

Bhattacharya and Tyagi (1990) mentioned that in some clinical studies dealing with cancer patients, the survival pattern follows the Rayleigh distribution. The probability density function (pdf) of survival time Y , following this distribution, is

$$f(y; \theta) = 2\theta y e^{-\theta y^2} \quad (1.1)$$

where $y > 0$ and $\theta > 0$. This distribution has a linearly increasing failure rate and this property makes it a suitable model for the lifetime of components that age rapidly with time (Johnson, Kotz and Balakrishnan, 1994, p. 457). The Rayleigh distribution is a special case of the well-known two-parameter Weibull distribution used as model for initiation of cancer (Lee, 1980, p. 162).

Corresponding to pdf (1.1) the survival distribution function, mean survival time and the hazard function are respectively given by

$$S(t) = e^{-\theta t^2}, \quad t > 0,$$

$$\mu = \frac{1}{2} \sqrt{\frac{\pi}{\theta}}$$

and

$$H(t) = 2\theta t, \quad t > 0.$$

The object of this paper is to obtain the Bayes estimators of $S(t)$, μ and $H(t)$ on the basis of the survival data recorded on n patients each with pdf (1.1). Suppose these patients are followed up in a clinical study and that the survival times of d of them are recorded and that $n - d$ of the survival times are censored and occur at random times. The ordered survival times recorded for the d deaths are denoted by $t_1, t_2, \dots, t_d$ ($0 < d < n$) and let the right censored times also be ordered and denoted by $t_{d+1}, t_{d+2}, \dots, t_n$. The

likelihood function (Dellaportas and Wright, 1991) conditional on θ is given by

$$l(t_1, \dots, t_n | \theta) = \left[\prod_{i=1}^d t_i \right] (2\theta)^d e^{-\theta T}, \quad (1.2)$$

where

$$T = \sum_{i=1}^d t_i^2 + \sum_{j=d+1}^{n-d} t_j^2.$$

It is also assumed here that some prior knowledge is available about the parameter θ of the Rayleigh model, based on past experience with similar survival data, and that this prior knowledge can be mathematically translated into suitable prior densities used in section 2 and section 3.

In Bayes estimation the loss function plays an important role. In this paper the linex loss function, introduced by Varian (1975), is used to derive the Bayes estimators. The linex loss function is useful when overestimation of the distribution parameters is more serious than their underestimation, or vice versa (Zellner, 1986).

The asymmetric linex loss function has the form

$$L(\Delta) = b(e^{a\Delta} - a\Delta - 1), \quad a \neq 0, b > 0 \quad (1.3)$$

where $\Delta = \hat{l}(\theta) - l(\theta)$ is the scalar estimation error of the parameter $l(\theta)$. The sign of the shape parameter a reflects the direction of the asymmetry and its magnitude reflects the degree of asymmetry. For small values of $|a|$, the function is almost symmetric and approximates a squared error loss function, which is also used in this paper. However, when $|a|$ assumes appreciable values, optimal point estimates and predictions will be quite different from those obtained with a squared error loss function (Zellner, 1986). Under the

linex loss function (1.3), the Bayes estimator of an integrable function $l(\theta)$ of θ , is defined as

$$\hat{l}(\theta) = -\frac{1}{a} \log E_{\theta} \left[e^{-al(\theta)} \right] \quad (1.4)$$

where E_{θ} stands for the posterior expectation.

2. Non-informative prior

For the situation where no prior knowledge about the parameter θ is available, consider the following improper pdf

$$g(\theta) \propto \frac{1}{\theta} \quad (2.1)$$

where $\theta > 0$, using Jeffreys' rule. The likelihood function (1.2) is combined with the prior pdf (2.1), via Bayes theorem, to obtain the posterior pdf

$$h(\theta | T, d) = \frac{T^d \theta^{d-1} e^{-\theta T}}{\Gamma(d)}, \quad \theta > 0. \quad (2.2)$$

The Bayes estimator of the hazard function, $H(t)$, under the linex loss function, is given by

$$\hat{H}_L(t) = \frac{d}{a} \log \left[1 + \frac{2at}{T} \right]. \quad (2.3)$$

Under squared error loss the Bayes estimator of $H(t)$ is

$$\hat{H}_S(t) = \frac{2td}{T}, \quad (2.4)$$

which is the same as the maximum likelihood estimator (MLE) from (1.2).

Remark 2.1

If one expands the right-hand side of (2.3) to $O(a^2)$, it is identical to the Bayes estimator of $H(t)$ under the squared error loss function, provided $-1 < \frac{2at}{T} \leq 1$.

Remark 2.2

The expansion of $\hat{H}_L(t)$ in terms of $\hat{H}_S(t)$

$$\hat{H}_L(t) = \hat{H}_S(t) - \frac{at}{T} \hat{H}_S(t) + \frac{4a^2t^2}{3T^2} \hat{H}_S(t) - \dots,$$

shows the direction of the deviation of $\hat{H}_L(t)$ from $\hat{H}_S(t)$ for small a . If T increases, the dependence on a becomes weaker, which can be regarded as a consistency property for a large sample size.

The Bayes estimator of mean survival time, μ , under linex loss from (1.4) is

$$\hat{\mu}_L = -\frac{1}{a} \log \left[\frac{T^d}{\Gamma(d)} \int_0^\infty \theta^{d-1} e^{-\theta T - \frac{a}{2} \sqrt{\frac{\pi}{\theta}}} d\theta \right].$$

By using the exponential series, it follows that

$$\hat{\mu}_L = -\frac{1}{a} \log \left[\sum_{k=0}^{2d-1} \frac{\left(\frac{-a\sqrt{\pi T}}{2}\right)^k \Gamma(d - \frac{k}{2})}{k! \Gamma(d)} \right], \quad (2.5)$$

and under the squared error loss function the Bayes estimator is simply

$$\hat{\mu}_S = \frac{\sqrt{\pi T} \Gamma(d - \frac{1}{2})}{2\Gamma(d)}, \quad (2.6)$$

which is the same if expanding the right-hand side of (2.5) to $O(a^2)$, provided $-2\Gamma(d) \leq a\sqrt{\pi T} \Gamma(d - \frac{1}{2}) < 2\Gamma(d)$.

The maximum likelihood estimator of μ from (1.2) is

$$\hat{\mu}_{MLE} = \sqrt{\frac{\pi T}{4d}}. \quad (2.7)$$

At any time t , the Bayes estimator under the linex loss function of $S(t)$ is

$$\hat{S}_L(t) = -\frac{1}{a} \log \left[\sum_{k=0}^{\infty} \frac{1}{k!} (-a)^k \left(1 + \frac{kt^2}{T}\right)^{-d} \right]. \quad (2.8)$$

Again, expanding the right-hand side of (2.8) to $O(a^2)$, gives the Bayes estimator under the squared error loss function

$$\hat{S}_S(t) = (1 + \frac{t^2}{T})^{-d}, \quad (2.9)$$

provided $-1 \leq a(1 + \frac{t^2}{T})^{-d} < 1$.

The maximum likelihood estimator is consequently

$$\hat{S}_{MLE}(t) = e^{-\frac{at^2}{T}}. \quad (2.10)$$

3. Conjugate prior

Suppose the parameter θ is characterized by the following conjugate pdf:

$$g(\theta | p, \beta) = \frac{1}{\Gamma(p)} \left(\frac{p}{\beta}\right)^p \theta^{p-1} e^{-\frac{p}{\beta}\theta} \quad (3.1)$$

where $\theta > 0$, $p > 0$ and $\beta > 0$, which is a gamma distribution with parameters $\frac{p}{\beta}$ and p .

Combining the likelihood function (1.2) with the prior pdf (3.1), via Bayes theorem, results in the posterior pdf

$$h(\theta | T, d, p, \beta) = \frac{1}{\Gamma(p+d)} \left(T + \frac{p}{\beta}\right)^{p+d-1} \theta^{p+d-1} e^{-\theta(T + \frac{p}{\beta})}, \quad \theta > 0 \quad (3.2)$$

which is again gamma distributed with parameters $T + \frac{p}{\beta}$ and $p + d$.

Remark 3.1

- The main reason for the popularity of this conjugate prior is the mathematical tractability of the gamma distribution.
- Reparameterize putting $\frac{\beta}{p} = \alpha$, in (3.1). Then the prior information embodied in (3.1) can be regarded as equivalent to having previously observed

p deaths and the sum of 'event' times α . This interpretation can be useful in choosing values of the hyperparameters.

3.1 Known hyperparameters p and β

The derivations under the squared error loss function are straightforward and the results are merely recorded:

$$\hat{H}_S(t) = \frac{2t(p+d)}{(T + \frac{p}{\beta})}, \quad (3.3)$$

$$\hat{\mu}_S = \frac{1}{2} \sqrt{\pi(T + \frac{p}{\beta})} \frac{\Gamma(p+d - \frac{1}{2})}{\Gamma(p+d)} \quad (3.4)$$

and

$$\hat{S}_S(t) = \left(\frac{T + \frac{p}{\beta}}{T + \frac{p}{\beta} + t^2} \right)^{p+d} \quad (3.5)$$

As the posterior pdf (3.2) is very similar to the posterior distribution in the non-informative case (2.2), the Bayes estimators of the three lifetime parameters, under the linex loss function, are of the same form, namely:

$$\hat{H}_L(t) = \frac{p+d}{a} \log \left[1 + \frac{2at}{T + \frac{p}{\beta}} \right], \quad (3.6)$$

$$\hat{\mu}_L = -\frac{1}{a} \log \left[\sum_{k=0}^{2(p+d)-1} \frac{(-a\sqrt{\pi})^k 2^{-k}}{k! \Gamma(p+d)} \left(T + \frac{p}{\beta} \right)^{\frac{k}{2}} \Gamma\left(p+d - \frac{k}{2}\right) \right], \quad (3.7)$$

and

$$\hat{S}_L(t) = -\frac{1}{a} \log \left[\sum_{k=0}^{\infty} \frac{1}{k!} (-a)^k \left(1 + \frac{kt^2}{T + \frac{p}{\beta}} \right)^{-(p+d)} \right]. \quad (3.8)$$

Expanding these estimators to $O(a^2)$, gives the Bayes estimators under the squared error loss function, provided $-(T + \frac{p}{\beta}) < 2at \leq (T + \frac{p}{\beta})$ for

(3.6), $-2\Gamma(p+d) \leq a\sqrt{\pi(T+\frac{p}{\beta})}\Gamma(p+d-\frac{1}{2}) < 2\Gamma(p+d)$ for (3.7) and $-1 \leq a(1+\frac{t^2}{T+\frac{p}{\beta}})^{-(p+d)} < 1$ for (3.8).

3.2 Prior distribution of β

In the previous section, it is straightforward to derive the Bayes estimators for the different lifetime distribution parameters in the case where the hyperparameters in the prior pdf are known. As an alternative procedure for unknown hyperparameters a hierarchical Bayes approach is considered.

Assume now that conditional on q , β follows the inverted gamma distribution with pdf:

$$g(\beta | q) = \frac{1}{\Gamma(q)} q^q \beta^{-(q+1)} e^{-\frac{q}{\beta}}, \quad (3.9)$$

where $\beta > 0$ and $q = 1, 2, \dots, n$. Here the hyperparameter q is chosen to reflect prior knowledge.

The posterior pdf, using Bayes theorem, is of the form

$$h(\theta, \beta | T, d, p, q) \propto l(t_1, \dots, t_d | \theta) g(\theta | p, \beta) g(\beta | q),$$

thus

$$h(\theta, \beta | T, d, p, q) \propto \frac{1}{\Gamma(p)} \frac{1}{\Gamma(q)} \left(\frac{p}{\beta}\right)^p q^q \beta^{-(q+1)} \theta^{d+p-1} e^{-\theta(T+\frac{p}{\beta})-\frac{q}{\beta}}. \quad (3.10)$$

3.2.1 Known hyperparameters p and q

Using the posterior pdf (3.10), the marginal posterior pdf of θ is

$$h(\theta | T, d, p, q) = C_1^{-1} \frac{\theta^{d+p-1} e^{-\theta T}}{(1 + \frac{p}{q}\theta)^{p+q}}, \quad (3.11)$$

where

$$C_1 = \left(\frac{q}{p}\right)^{p+d} \Gamma(p+d) \Psi(p+d, d-q+1, \frac{q}{p}T)$$

and $\Psi(\cdot)$ is the type II confluent hypergeometric function introduced by Tricomi (Erdélyi et al., 1953, p. 182).

Remark 3.2

This posterior pdf (3.11) is identical to the case where it can be assumed that the prior knowledge about θ follows an F-distribution with p and q degrees of freedom respectively.

In this case, assume now that the hyperparameters p and q in (3.11) are in fact known degrees of freedom (i.e. discrete values). This F prior has the advantage that the hyperparameters are discrete, which reduces the hyperparameter space, and that the parameter q is the only parameter that is important for at least the first moment of the F-distribution.

The Bayes estimators for the hazard function $H(t)$, under the squared error loss function and linex loss function, are respectively

$$\hat{H}_S(t) = \frac{2tq(p+d)\Psi(p+d+1, d-q+2, \frac{q}{p}T)}{p\Psi(p+d, d-q+1, \frac{q}{p}T)} \quad (3.12)$$

and

$$\hat{H}_L(t) = -\frac{1}{a} \log \left[\frac{\Psi(p+d, d-q+1, \frac{q}{p}(T+2at))}{\Psi(p+d, d-q+1, \frac{q}{p}T)} \right] \quad (3.13)$$

The Bayes estimator under the linex loss function for μ , the mean survival time, is

$$\hat{\mu}_L = -\frac{1}{a} \log \left[\sum_{k=0}^{2(p+d)-1} \frac{1}{k!} \left(-\frac{a}{2} \sqrt{\frac{p\pi}{q}} \right)^k \times \frac{\Gamma(p+d-\frac{k}{2}) \Psi(p+d-\frac{k}{2}, d-q-\frac{k}{2}+1, \frac{q}{p}T)}{\Gamma(p+d) \Psi(p+d, d-q+1, \frac{q}{p}T)} \right] \quad (3.14)$$

Expansion of the right hand side of (3.14) to $O(a^2)$ gives the Bayes estimator under the squared error loss function,

$$\hat{\mu}_S = \sqrt{\frac{p\pi}{4q}} \frac{\Gamma(p+d-\frac{1}{2}) \Psi(p+d-\frac{1}{2}, d-q+\frac{1}{2}, \frac{q}{p}T)}{\Gamma(p+d) \Psi(p+d, d-q+1, \frac{q}{p}T)}, \quad (3.15)$$

provided $-\Gamma(p+d) \Psi(p+d, d-q+1, \frac{q}{p}T) \leq \frac{a}{2} \sqrt{\frac{p\pi}{q}} \Gamma(p+d-\frac{1}{2}) \times \Psi(p+d-\frac{1}{2}, d-q+\frac{1}{2}, \frac{q}{p}T) < \Gamma(p+d) \Psi(p+d, d-q+1, \frac{q}{p}T)$.

The Bayes estimator for the survival distribution function $S(t)$ under the linex loss function is

$$\hat{S}_L(t) = -\frac{1}{a} \log \left[\sum_{k=0}^{\infty} \frac{(-a)^k \Psi(p+d, d-q+1, \frac{q}{p}(T+kt^2))}{k! \Psi(p+d, d-q+1, \frac{q}{p}T)} \right] \quad (3.16)$$

Expanding the right-hand side of (3.16) to $O(a^2)$, gives the Bayes estimator under the squared error loss function

$$\hat{S}_S(t) = \frac{\Psi(p+d, d-q+1, \frac{q}{p}(T+t^2))}{\Psi(p+d, d-q+1, \frac{q}{p}T)}, \quad (3.17)$$

provided $-\Psi(p+d, d-q+1, \frac{q}{p}T) \leq a \Psi(p+d, d-q+1, \frac{q}{p}(T+t^2)) < \Psi(p+d, d-q+1, \frac{q}{p}T)$.

3.2.2 Unknown hyperparameters p and q

From the previous sections it can be seen that for known hyperparameters of the prior densities, the estimators for the lifetime distribution parameters can be obtained either in closed form or by means of an approximation. The researcher of survival times is usually confronted with the problem of what values to assign to these hyperparameters, if they are unknown.

Consider once again the situation where the prior knowledge about θ follows the gamma pdf (3.1) and the random hyperparameter β follows an inverted gamma pdf (3.9). Further, let the hyperparameters p and q in this compound posterior distribution (3.10) be unknown. A method used by Soland (1969) is paralleled here. Suppose both degrees of freedom (see remark 3.2) p and q are restricted to a finite number of values $p_1, \dots, p_{m_1}$ and $q_1, \dots, q_{m_2}$ respectively. Thus assume the following known mass functions for p and q as follows:

$$\Pr(p = p_i) = \tau_i, \quad \sum_{i=1}^{m_1} \tau_i = 1 \quad \text{for} \quad i = 1, \dots, m_1$$

and

$$\Pr(q = q_j) = \zeta_j, \quad \sum_{j=1}^{m_2} \zeta_j = 1 \quad \text{for} \quad j = 1, \dots, m_2.$$

If no information exists on what values to use for these mass functions, one can assume that the hyperparameters p and q lie uniformly in given intervals. The joint posterior pdf (3.10) for unknown p and q becomes

$$h(\theta, \beta, p_i, q_j | T, d) \propto \frac{1}{\Gamma(p_i)} \frac{1}{\Gamma(q_j)} \left(\frac{p_i}{\beta} \right)^{p_i} q_j^{q_j} \beta^{-(q_j+1)} \theta^{d+p_i-1} \times e^{-\theta(T + \frac{p_i}{\beta}) - \frac{q_j}{\beta} \tau_i \zeta_j}. \quad (3.18)$$

The marginal posterior pdf for θ using (3.18) is simply

$$h(\theta | T, d) = C_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{\theta^{d+p_i-1} e^{-\theta T (\frac{p_i}{q_j})^{p_i} \tau_i \zeta_j}}{B(p_i, q_j) (1 + \frac{p_i}{q_j} \theta)^{p_i+q_j}}, \quad i = 1, \dots, m_1$$

and $j = 1, \dots, m_2$; (3.19)

where

$$C_2 = \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{1}{B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j} \right)^{-d} \Gamma(p_i + d) \Psi(p_i + d, d - q_j + 1, \frac{q_j}{p_i} T).$$

Table 3.1 summarizes the different Bayes estimators for the lifetime distribution parameters.

Table 3.1. Estimators for the lifetime distribution parameters

Lifetime parameter	Estimator
$H_S(t)$	$2tC_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{1}{B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j}\right)^{-d-1} \Gamma(p_i + d + 1)$ $\times \Psi(p_i + d + 1, d - q_j + 2, \frac{q_i}{p_i} T)$
$H_L(t)$	$-\frac{1}{a} \log[C_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{1}{B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j}\right)^{-d} \Gamma(p_i + d)$ $\times \Psi(p_i + d, d - q_j + 1, \frac{q_i}{p_i} (T + 2at))]$
μ_S	$C_2^{-1} \frac{\sqrt{\pi}}{2} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{1}{B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j}\right)^{-d+\frac{1}{2}} \Gamma(p_i + d - \frac{1}{2})$ $\times \Psi(p_i + d - \frac{1}{2}, d - q_j + \frac{1}{2}, \frac{q_i}{p_i} T)$
μ_L	$-\frac{1}{a} \log[C_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \sum_{k=0}^{2(\min(p_i)+d)-1} \frac{(-\frac{a\sqrt{\pi}}{2})^k}{k! B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j}\right)^{-d+\frac{k}{2}}$ $\times \Gamma(p_i + d - \frac{k}{2}) \Psi(p_i + d - \frac{k}{2}, d - q_j - \frac{k}{2} + 1, \frac{q_i}{p_i} T)]$
$S_S(t)$	$C_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{1}{B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j}\right)^{-d} \Gamma(p_i + d)$ $\times \Psi(p_i + d, d - q_j + 1, \frac{q_i}{p_i} (T + t^2))$
$S_L(t)$	$-\frac{1}{a} \log[C_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \sum_{k=0}^{\infty} \frac{(-a)^k}{k! B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j}\right)^{-d} \Gamma(p_i + d)$ $\times \Psi(p_i + d, d - q_j + 1, \frac{q_i}{p_i} (T + kt^2))]$

Lastly, this theory is now extended to the case where one wishes to predict a future patient's lifetime, say, $y > t_d$. Assume now that the same prior knowledge about θ exists, namely the F prior distribution, with unknown degrees of freedom p and q . Let $z = y - t_d$. Given the same type of sample, the conditional joint pdf of z and θ , is

$$h(z, \theta | T, d) = h(\theta | T, d) f(z | \theta) \quad (3.20)$$

where $h(\theta | T, d)$ is the marginal posterior pdf of θ (3.19) and $f(z | \theta)$ is the pdf from a Rayleigh distribution (1.1).

With p and q unknown and using (3.20), the predictive posterior pdf of z is simply

$$h(z | T, d) = C_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{2z}{B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j} \right)^{-d-1} \Gamma(p_i + d + 1) \\ \times \Psi(p_i + d + 1, d - q_j + 2, \frac{q_j}{p_i} (T + z^2)) \quad (3.21)$$

where C_2 is defined as in (3.19), since

$$C_2 = \int_0^\infty \int_0^\infty h(\theta | T, d) f(z | \theta) d\theta dz.$$

Using

$$h(z | T, d) = \int_0^\infty h(z, \theta | T, d) d\theta$$

the Bayes predictor of z under the squared error loss function becomes

$$\hat{z}_S = C_2^{-1} \frac{\sqrt{\pi}}{2} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \frac{\tau_i \zeta_j}{B(p_i, q_j)} \left(\frac{p_i}{q_j} \right)^{-d+\frac{1}{2}} \Gamma(p_i + d - \frac{1}{2}) \\ \times \Psi(p_i + d - \frac{1}{2}, d - q_j + \frac{1}{2}, \frac{q_j}{p_i} T), \quad (3.22)$$

which is the same as the Bayes estimator of μ under the squared error loss function (see table 3.1).

The Bayes predictor of a future observation under the linex loss function is

$$\begin{aligned}\hat{z}_L &= -\frac{1}{a} \log \left[\int_0^{\infty} e^{-az} h(z \mid T, d) dz \right] \\ &= -\frac{1}{a} \log \left[\int_0^{\infty} \int_0^{\infty} e^{-az} C_2^{-1} h(z, \theta \mid T, d) d\theta dz \right]\end{aligned}$$

which after using the exponential series, reduces to

$$\begin{aligned}\hat{z}_L &= -\frac{1}{a} \log \left[C_2^{-1} \sum_{i=1}^{m_1} \sum_{j=1}^{m_2} \sum_{k=0}^{2(\min(p_i)+d)-1} \frac{(-a)^k \Gamma(\frac{k}{2}+1)}{k! B(p_i, q_j)} \tau_i \zeta_j \left(\frac{p_i}{q_j}\right)^{-d+\frac{k}{2}} \right. \\ &\quad \left. \times \Gamma(p_i + d - \frac{k}{2}) \Psi(p_i + d - \frac{k}{2}, d - q_j - \frac{k}{2} + 1, \frac{q_j}{p_i} T) \right].\end{aligned}\tag{3.23}$$

4. Practical example

To illustrate the application of the results to the analysis of survival data, consider the survival times in years of a group of patients given chemotherapy and radiation treatment shown in table 4.1; this is a subset of data reported by Stablein et al. (1981).

Table 4.1. Survival times in years of gastric cancer patients

Chemotherapy and radiation		
0,047	0,507	1,485
0,115	0,529	1,553
0,121	0,534	1,581
0,132	0,540	1,589
0,164	0,570	2,178
0,197	0,641	2,343
0,203	0,644	2,416*
0,260	0,696	2,444*
0,282	0,841	2,825*
0,296	0,863	2,830*
0,334	1,099	3,578*
0,395	1,219	3,658*
0,458	1,271	3,743
0,466	1,326	3,978*
0,501	1,447	4,033*

* Indicates censored observations.

Figure 4.1 shows the Kaplan-Meier estimator (KME) of $F(t)$, (Kaplan and Meier, 1958), constructed from this sample and the distribution function of the Rayleigh pdf fitted by the method of maximum likelihood. The 95% confidence limits for the KME are also shown in figure 4.1. The graphs indicate that the Rayleigh model is acceptable for these data, which entails assuming that the instantaneous risk of failure is linear over time.

The linex loss function, with various values of the parameter α , is used to estimate the mean survival time, the hazard function and the survival

distribution function. The tendency to increase or decrease estimates, relative to squared error loss estimates, can be evaluated, thus providing guidelines for choice of a value of α .

Applying the Rayleigh model (1.1) to the data in table 4.1, the value of

$$T = \sum_{i=1}^d t_i^2 + \sum_{j=d+1}^{n-d} t_j^2 = 133,643, \quad \text{with } d = 37.$$

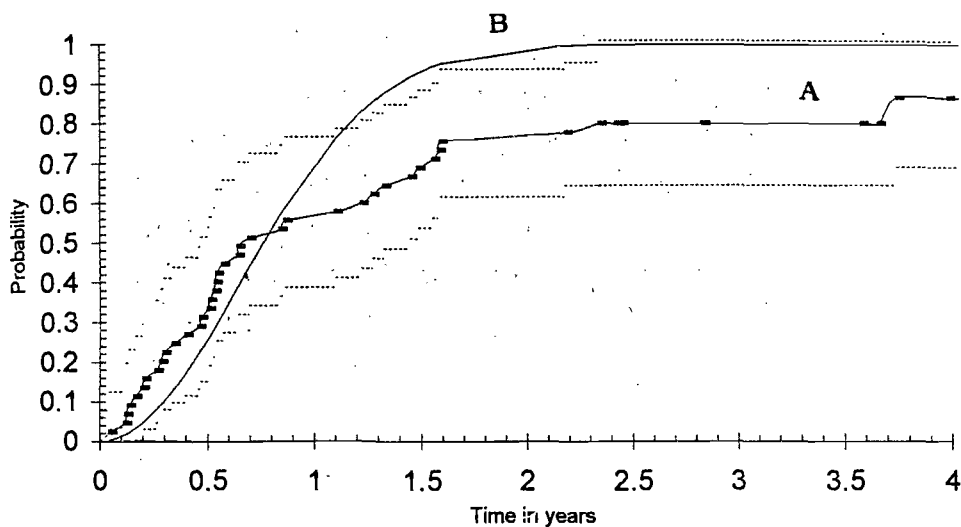


Figure 4.1. Kaplan-Meier distribution function

A: KME of $F(t)$ B: $F(t)$ of pdf(1.1)

4.1 Non-informative prior

Estimates of the mean survival time, μ , are given in table 4.2.

Table 4.2. Estimates of the mean survival time for the non-informative prior

$\hat{\mu}_{MLE}$ (2.7)	$\hat{\mu}_S$ (2.6)	$\hat{\mu}_L(2.5)$				
		$a = 1$	$a = 0,5$	$a = -1,5$	$a = -5$	$a = -15$
1,684	1,702	1,700	1,701	1,705	1,712	1,734

For small $|a|$, the estimates of μ under the linex loss function are very close to the estimate of μ under the squared error loss function. This is one of the useful properties of working with the linex loss function. Also, if $a < 0$, one can see that μ is overestimated and when $a > 0$, μ is underestimated. The effect of a in this case is quite small, due to a relative large sample size.

Figure 4.2 gives the plot of the hazard function $H(t)$ for the non-informative prior.

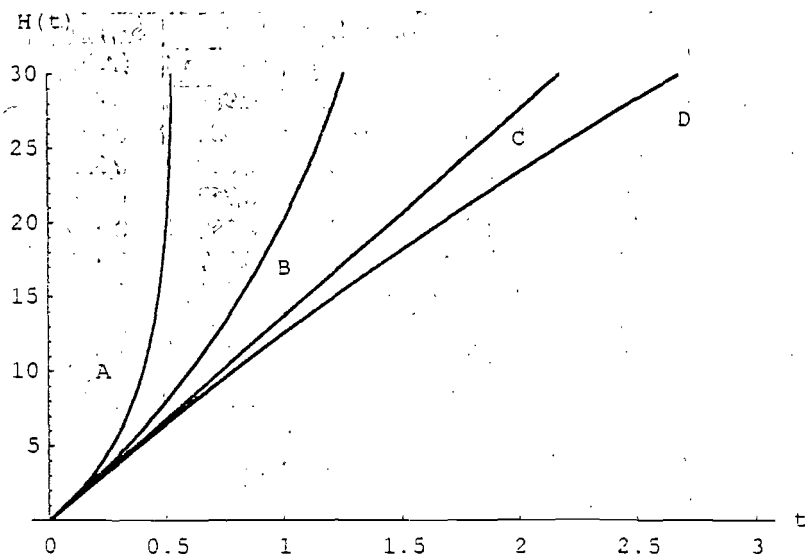


Figure 4.2. Hazard function using non-informative prior

- A: $\hat{H}_L(t); a = -5; (2.3)$ C: $\hat{H}_S(t); \hat{H}_{MLE}(t); (2.4)$
 B: $\hat{H}_L(t); a = -1, 5; (2.3)$ D: $\hat{H}_L(t); a = 0, 5; (2.3)$

One can clearly see the over-/underestimation of the hazard function at any time t for different values of a .

In table 4.3 the probability is given of surviving a specific time, t , for gastric cancer patients who are treated with chemotherapy and radiation.

Table 4.3. Estimates of the survival distribution function for the non-informative prior

t	$\hat{S}_S(t)(2.9)$	$\hat{S}_L(t)(2.8)$			$\hat{S}_{MLE}(t)$ (2.10)
		$a = 0,5$	$a = -1,5$	$a = -5$	
0,25	0,983	0,983	0,983	0,983	0,983
0,50	0,933	0,933	0,933	0,934	0,933
0,75	0,856	0,856	0,856	0,857	0,856
1,00	0,759	0,759	0,760	0,762	0,758
1,25	0,651	0,650	0,652	0,656	0,649
1,50	0,539	0,538	0,541	0,547	0,536
1,75	0,432	0,432	0,435	0,441	0,428
2,00	0,336	0,335	0,339	0,345	0,330
2,25	0,253	0,252	0,255	0,261	0,246
2,50	0,184	0,184	0,186	0,191	0,177

The above illustration shows that estimation is fairly straightforward, given no information about θ .

4.2 Informative prior

Assume now that information about θ does exist and θ is distributed according to the F-distribution as discussed in section 3.2.2, where the hyperparameters of the F prior, p and q , are unknown. Assume now that the hyperparameters p and q lie uniformly in the intervals $(0;5]$ and $(1;5]$ respectively (Evans and Nigm, 1980). Upper limit values greater than five have little influence on the results and computational times are drastically reduced with intervals of which the upper limits are equal to five. The mass functions are summarized in tables 4.4 and 4.5 for p and q .

Table 4.4. Mass function for p

p	1	2	3	4	5
mass	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$

Table 4.5. Mass function for q

q	2	3	4	5
mass	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

The values for estimating μ from table 3.1 are given in table 4.6, for the F prior where p and q are unknown.

Table 4.6. Estimates of the mean survival time for the F prior, where p and q are unknown

$\hat{\mu}_S$	$\hat{\mu}_L$			
	$a = 0,5$	$a = -1,5$	$a = -5$	$a = -15$
1,647	1,627	1,690	1,734	1,781

The estimates for $H(t)$ and $S(t)$ are given in table 4.7 for the F prior where p and q are unknown.

Table 4.7. Estimates of $H(t)$ and $S(t)$ for the F prior, where p and q are unknown

t	$\hat{H}_S(t)$	$\hat{H}_L(t)$			$\hat{S}_S(t)$	$\hat{S}_L(t)$		
		a				a		
		0,5	-1,5	-5,0		0,5	-1,5	-5,0
0,25	0,655	0,652	0,664	0,687	0,984	0,984	0,984	0,984
0,75	1,964	1,937	2,051	2,310	0,863	0,863	0,864	0,864
1,25	3,273	3,200	3,554	4,411	0,666	0,665	0,667	0,671
1,75	4,583	4,440	5,126	7,429	0,452	0,452	0,455	0,461
2,25	5,892	5,660	6,811	12,875	0,272	0,271	0,275	0,281

For unknown p and q , this example has shown how one can realistically derive estimators for the parameters in the Rayleigh model. This has also shown that the estimation of the parameters produced far better results than the estimation in the non-informative case.

Lastly, the example is extended to the case where the prediction of the lifetime of a future patient is of interest. Table 4.8 summarizes the Bayes predictions of the lifetime z of a future patient. The same F prior with unknown p and q are used as illustration.

Table 4.8. Bayes prediction of the lifetime for a future patient

$\hat{z}_S$	$\hat{z}_L$			
	$a = 0,5$	$a = -1,5$	$a = -5$	$a = -15$
1,647	1,585	1,825	2,268	4,279

5. Concluding remarks

The results of this paper give guidelines for deriving estimators of the distribution parameters: mean survival time, hazard function and the survival distribution function. If the Rayleigh model is appropriate for modelling the lifetimes of cancer patients, the paper has shown how to implement prior knowledge of the parameter θ using the linex loss function. The analytical ease with which results can be obtained using the asymmetric linex loss function makes it attractive for use in applied problems and in assessing the effects of departures from an assumed asymmetric loss function.

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